23CS18E FULL STACK APP WITH FLUTTER AND FIREBASE

MESS MANAGEMENT SYSTEM

TEAM MEMBERS

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CHAPTER 1

INTRODUCTION

The management of food services in college hostels is a critical yet often overlooked aspect of student welfare. Traditional mess systems rely heavily on manual processes — paper-based complaints, verbal feedback, handwritten menus, and offline billing — resulting in inefficiencies, communication gaps, and student dissatisfaction. There is a growing need for a digital platform that streamlines mess operations, gives students a voice, and provides administrators with real-time control over the entire food service system.

To address these challenges, the Mess Management System is developed as a comprehensive digital platform for hostel mess operations. The system enables students to view the daily menu with nutritional information, rate individual dishes, file food quality complaints, suggest new items, declare meal intent, and request additional meals. It supports real-time feedback and transparent tracking of all interactions between students and mess administration.

Simultaneously, the application provides administrative functionalities such as daily menu management, complaint handling with severity levels, suggestion review with voting, monthly billing generation, and push notification delivery to students. Wardens receive role-specific access to hostel data, while administrators manage the entire ecosystem. The system integrates students, wardens, and administrators into a unified real-time environment.

The application is developed using Flutter for cross-platform mobile and web development, Firebase Authentication for secure login, Cloud Firestore for real-time database management, and Provider for state management. Local data caching is handled through SharedPreferences and SQLite for offline support. Firebase Cloud Messaging enables push notifications, while a Node.js server handles FCM dispatch. Campus navigation features are also integrated with Google Maps, GPS, and voice-based assistance.

1.1 OBJECTIVES

- To develop a user-friendly mobile application for hostel mess management
- To enable students to view, rate, and give feedback on daily meals
- To provide a structured complaint submission and resolution system with severity levels
- To allow dish suggestions with community voting and admin approval workflow
- To support meal intent declaration for accurate mess planning
- To manage additional food requests outside regular meal schedules
- To automate monthly mess billing with payment tracking
- To provide role-based access for Students, Admins, and Wardens
- To deliver real-time push notifications via Firebase Cloud Messaging
- To integrate campus navigation with Google Maps and GPS support
- To support offline functionality through local caching using SharedPreferences and SQLite
- To design a scalable and secure database using Cloud Firestore

CHAPTER 2

DATABASE DESIGN

The Mess Management System uses Cloud Firestore as its primary database, which follows a collection-based NoSQL structure. This design ensures flexibility, scalability, and real-time data synchronization across all platforms — Android, iOS, Web, and Desktop.

Each component such as users, menus, ratings, complaints, suggestions, meal intents, billing, and additional food requests is maintained in separate Firestore collections. This modular structure improves data organization and system maintainability.

The system implements role-based access control, where:

- Students manage their own ratings, complaints, and meal intents
- Wardens access hostel-specific data and reports
- Admins control the entire system including menus, billing, and complaints

Local caching using SharedPreferences and SQLite supplements Firestore with offline support, storing menus, ratings, complaints, and session data on the device.

2.1 DATABASE STRUCTURE

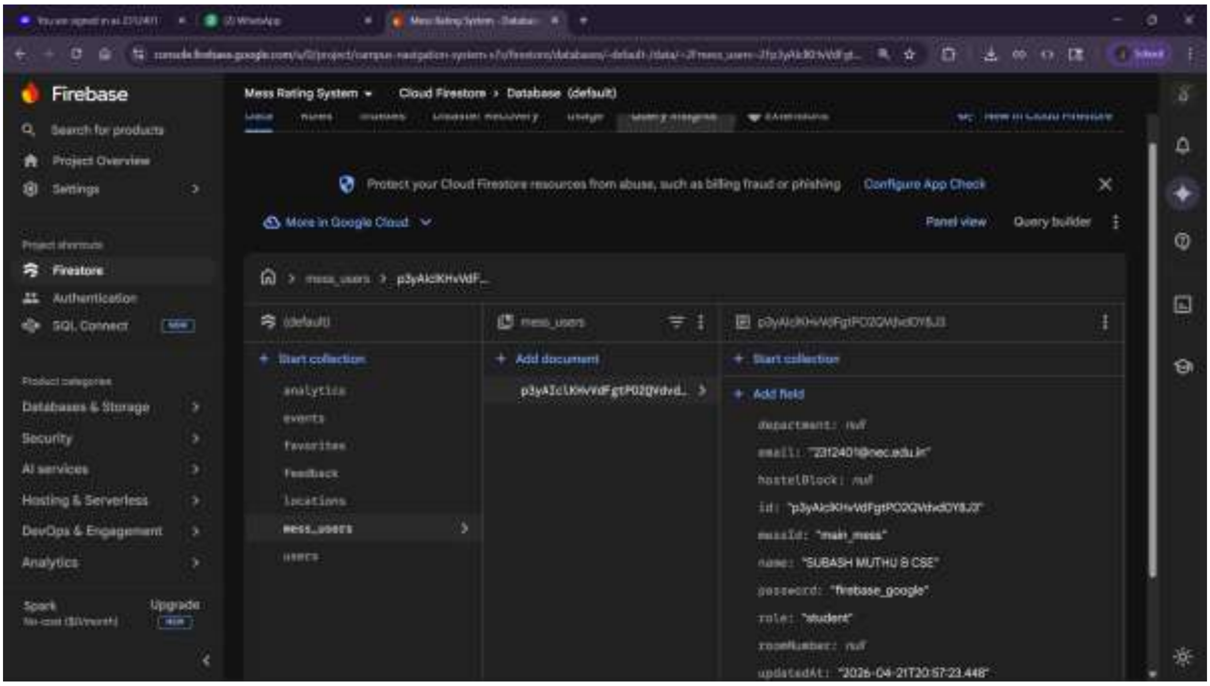

Fig. 2.1 — Database Overview

Figure 2.1 shows the overall Cloud Firestore database structure of the Mess Management System, illustrating how different collections for users, menus, ratings, complaints, suggestions, billing, and campus data are organized for efficient data management.

mess_users Collection Structure

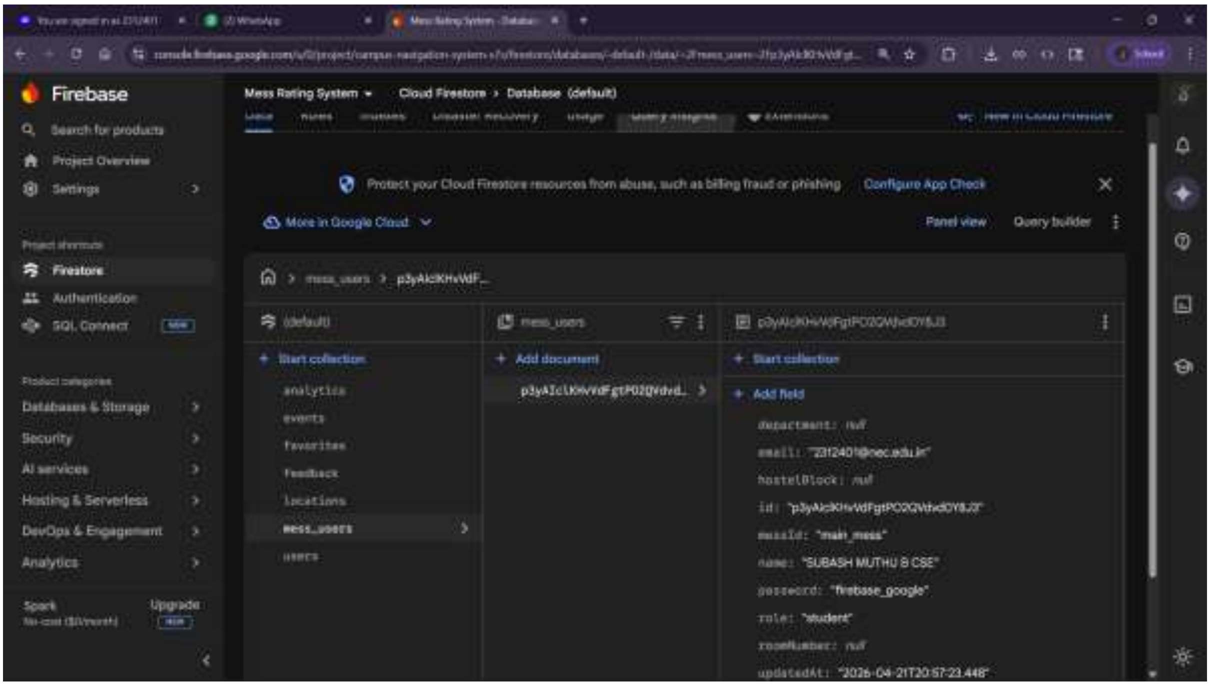

Fig. 2.2 — mess_users Collection Structure

Figure 2.2 illustrates the structure of the mess_users collection, which stores user details such as name, email, role, hostel block, room number, and profile information for role-based access control.

```
mess_users
└── userId
    ├── name
    ├── email
    ├── role          (student / admin / warden)
    ├── hostelBlock
    ├── roomNumber
```

├── department

├── year

├── profileImageUrl

└── createdAt

mess_menus Collection Structure

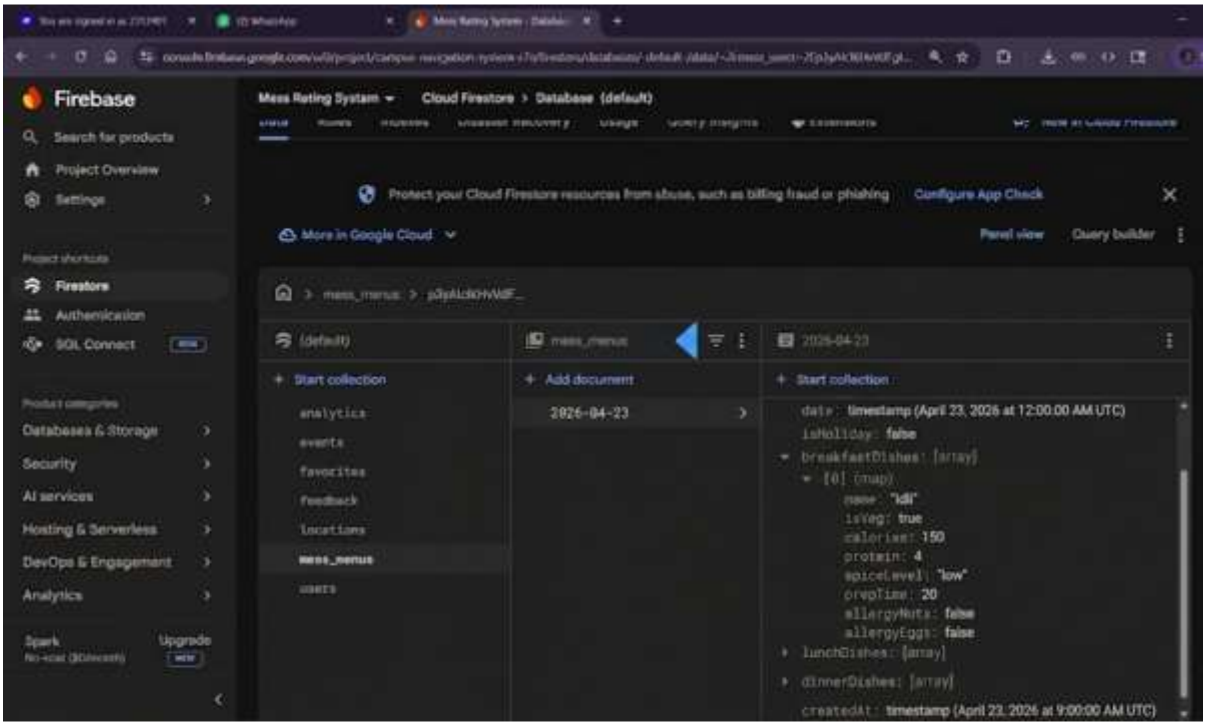

Fig. 2.3 — mess_menus Collection Structure

Figure 2.3 shows the mess_menus collection structure that maintains daily menu entries including breakfast, lunch, and dinner dishes, along with holiday flags and menu history for each date.

mess_menus

└── dateKey (e.g., "2026-04-23")

├── date

├── isHoliday

```
├── breakfastDishes[]
├── lunchDishes[]
├── dinnerDishes[]
└── createdAt
```

Dish (embedded object):

```
├── name
├── isVeg            (true / false)
├── calories
├── protein
├── spiceLevel
├── prepTime
├── allergyNuts
└── allergyEggs
```

mess_ratings Collection Structure

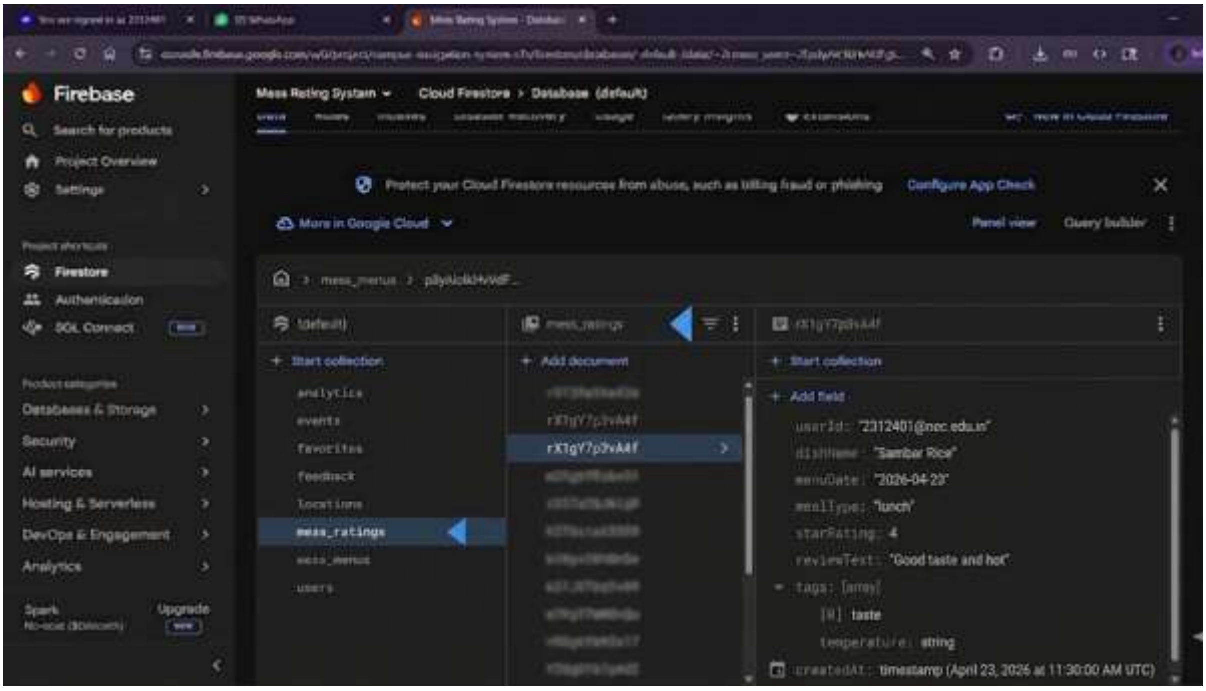

Fig. 2.4 — mess_ratings Collection Structure

Figure 2.4 illustrates the mess_ratings collection, which stores individual dish ratings submitted by students including star ratings, review text, and tag-based feedback for quality analysis.

mess_ratings

└── ratingId

├── userId

├── dishName

├── menuDate

├── mealType (breakfast / lunch / dinner)

├── starRating (1–5)

├── reviewText

├── tags[] (e.g., taste, quality, temperature)

└── createdAt

mess_complaints Collection Structure

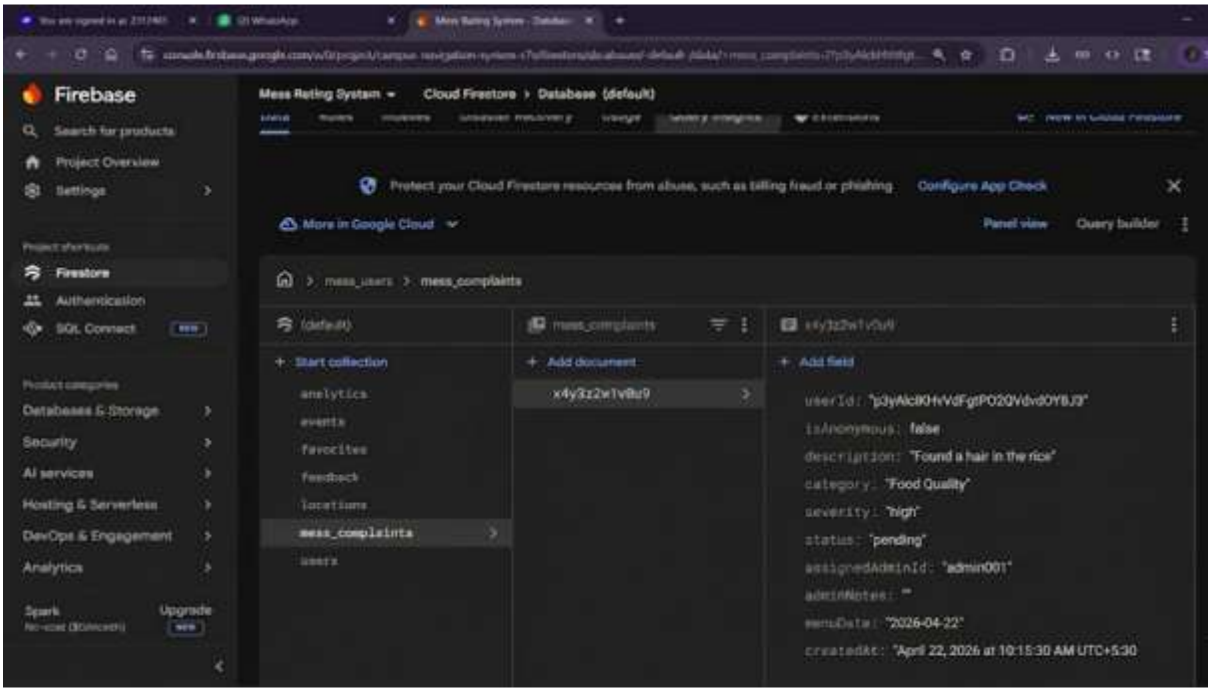

Fig. 2.5 — mess_complaints Collection Structure

Figure 2.5 shows the mess_complaints collection structure, which manages food quality complaints submitted by students with severity classification, anonymous submission support, admin assignment, and status tracking.

mess_complaints

└── complaintId

├── userId

├── isAnonymous

├── description

├── category

├── severity (low / medium / high / critical)

├── status (pending / under-review / resolved)

├── assignedAdminId

├── adminNotes

├── menuDate

└── createdAt

mess_suggestions Collection Structure

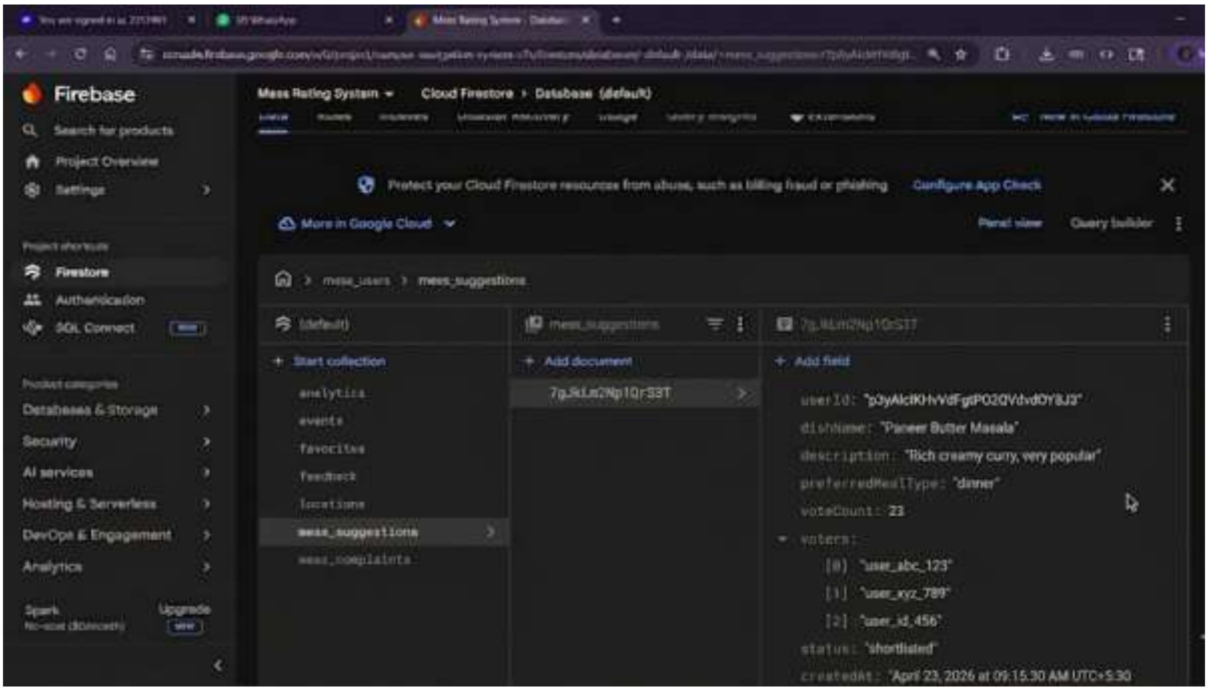

Fig. 2.6 — mess_suggestions Collection Structure

Figure 2.6 illustrates the mess_suggestions collection that stores dish suggestions proposed by students, including community vote counts and a status workflow from proposal through to scheduling.

mess_suggestions

└── suggestionId

├── userId

├── dishName

├── description

├── preferredMealType

├── voteCount

├── voters[]

├── status (proposed / shortlisted / accepted / scheduled)

└── createdAt

mess_bills Collection Structure

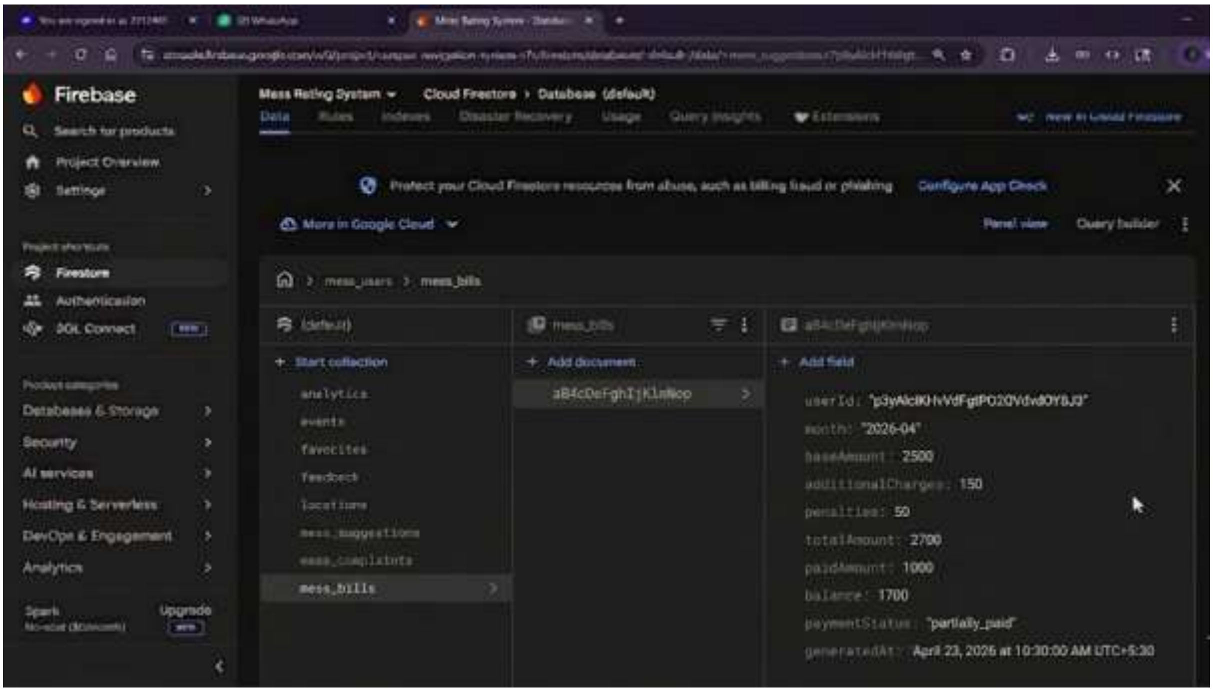

Fig. 2.7 — mess_bills Collection Structure

Figure 2.7 shows the mess_bills collection, which maintains monthly billing records for each student including base amount, additional charges, penalties, payment status, and balance calculation.

mess_bills

└── billId

├── userId

├── month (e.g., "2026-04")

├── baseAmount

├── additionalCharges

├── penalties

├── totalAmount

├── paidAmount

├── balance

├── paymentStatus (unpaid / partially_paid / paid)

└── generatedAt

additional_food_requests Collection Structure

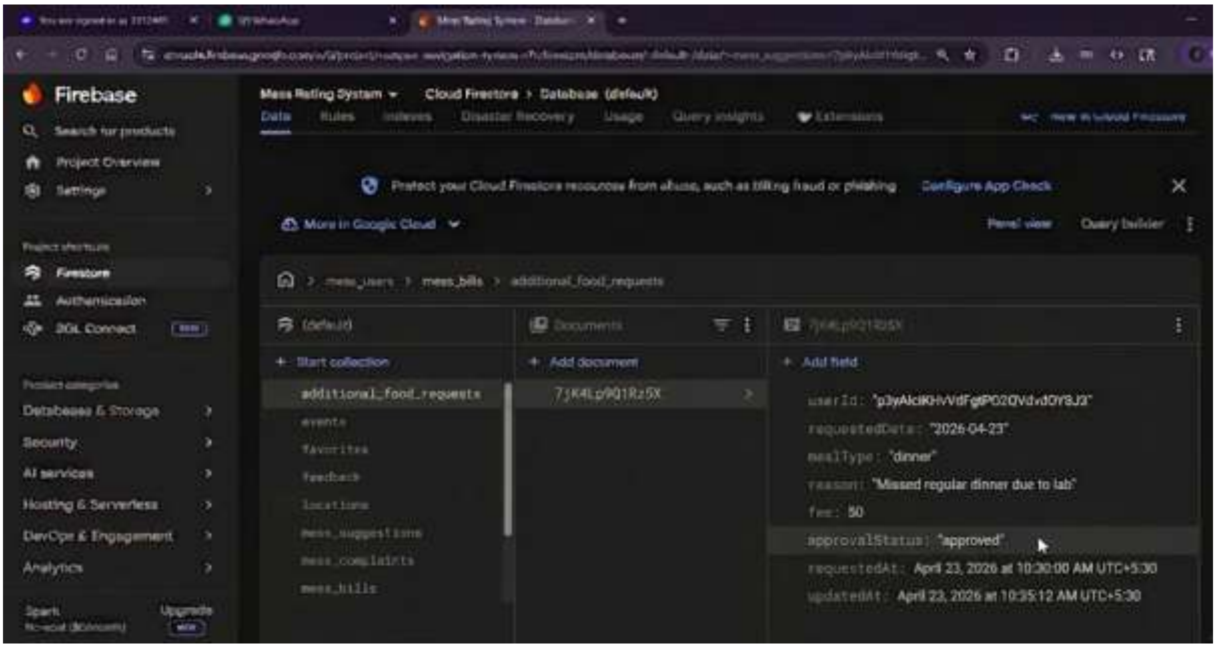

Fig. 2.8 — additional_food_requests Collection Structure

Figure 2.8 illustrates the additional_food_requests collection that stores requests made by students for extra meals outside the regular schedule, including approval status, time window enforcement, and fee calculation.

```
additional_food_requests
└── requestId
    ├── userId
    ├── requestedDate
    ├── mealType
    ├── reason
    ├── fee
    ├── approvalStatus   (pending / approved / rejected)
    ├── requestedAt
    └── updatedAt
```

CHAPTER 3

IMPLEMENTATION AND RESULTS

The Mess Management System is implemented using a modular architecture. Each function such as authentication, menu management, complaint handling, rating, suggestion voting, and billing is developed as a separate module. Firebase ensures real-time synchronization of all operations across Student, Admin, and Warden roles.

3.1 Authentication Module

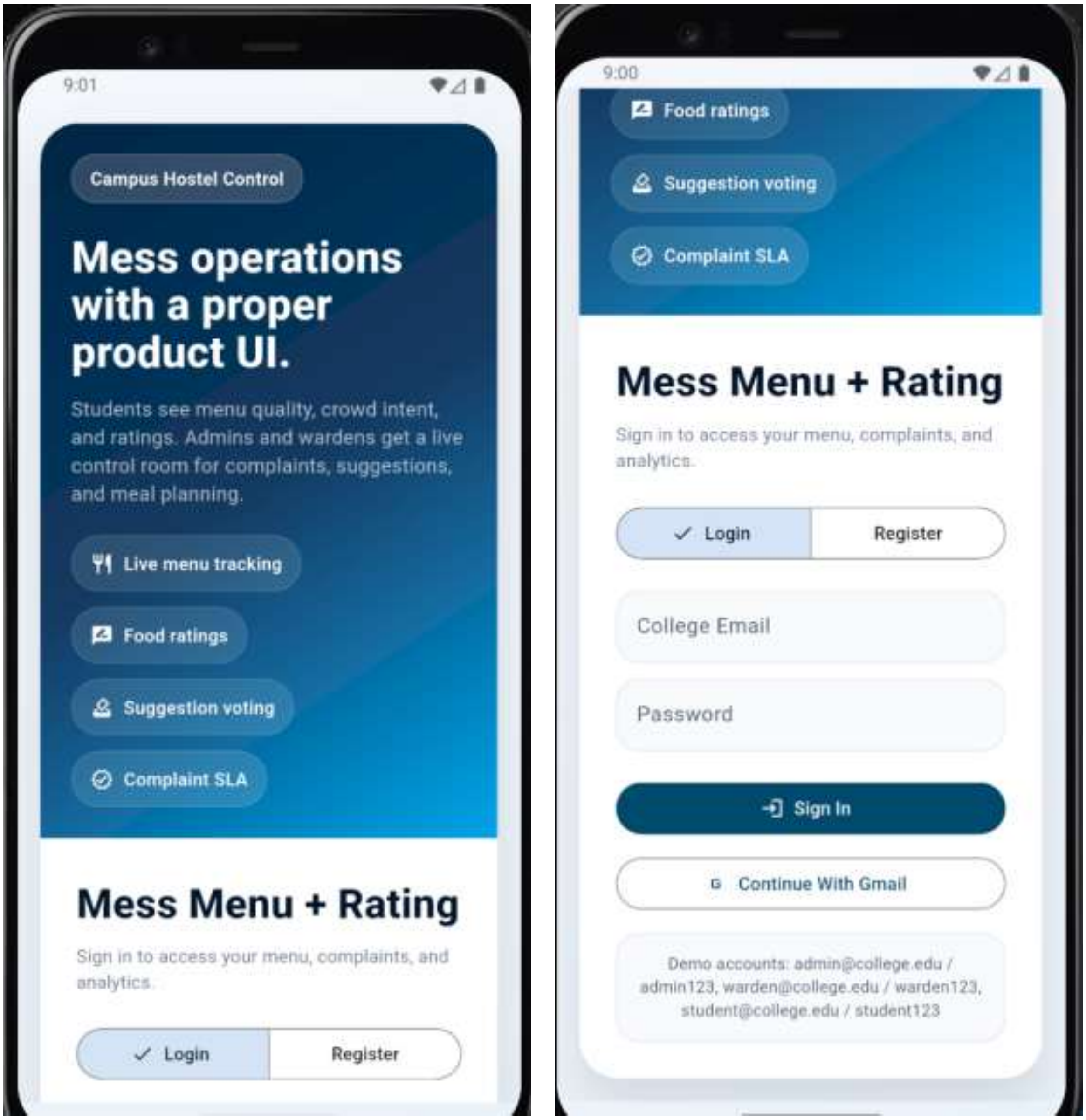

Fig. 3.1.1 — Login & Signup Page

Figure 3.1.1 illustrates the authentication interface of the Mess Management System, allowing students, wardens, and admins to register and log in securely using Firebase Authentication with email/password and Google Sign-In support.

3.2 Admin Module

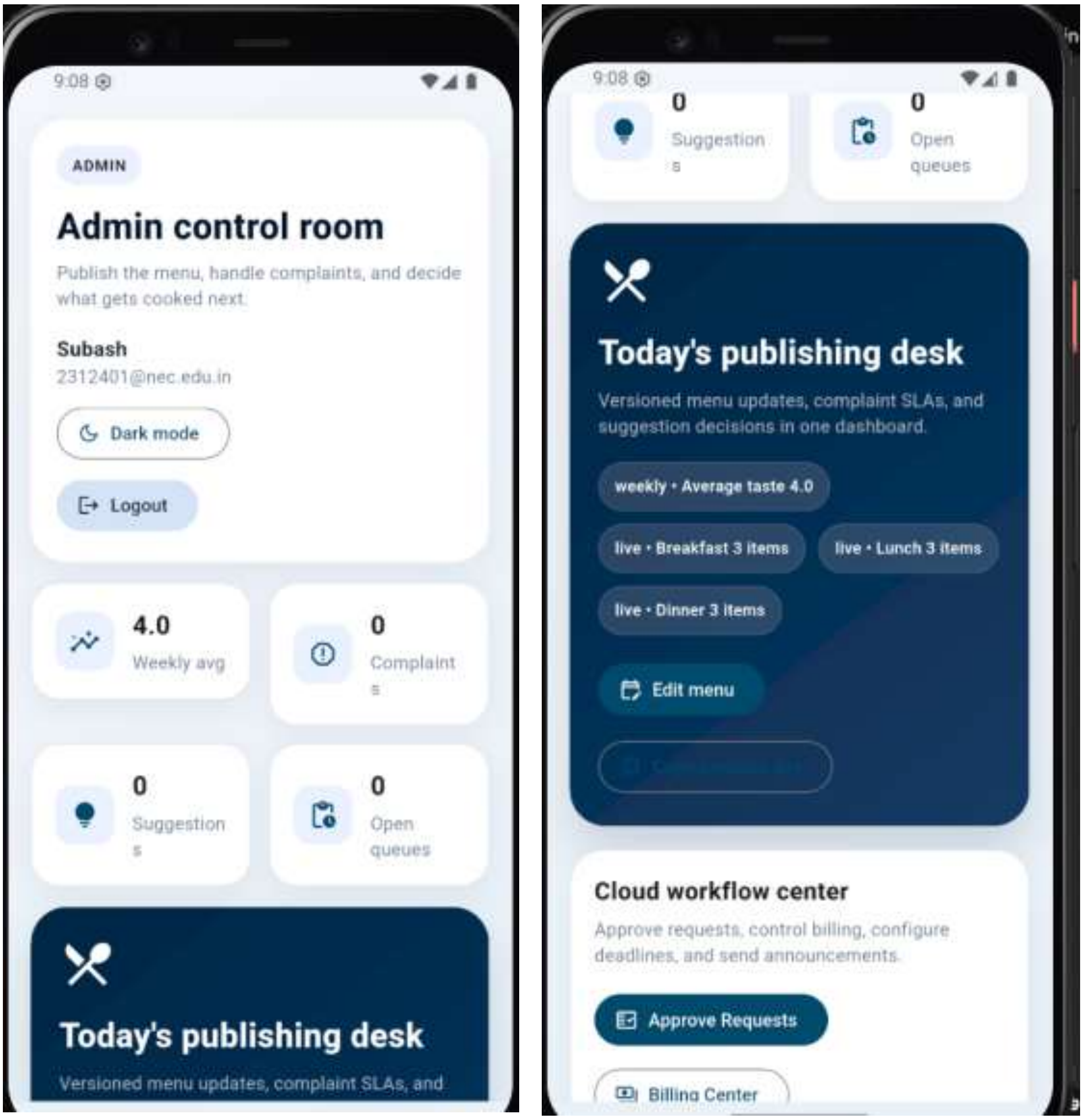

Fig. 3.2.1 — Admin Dashboard

Figure 3.2.1 shows the admin dashboard providing a consolidated overview of pending complaints, active menu status, recent suggestions, and system announcements for daily monitoring.

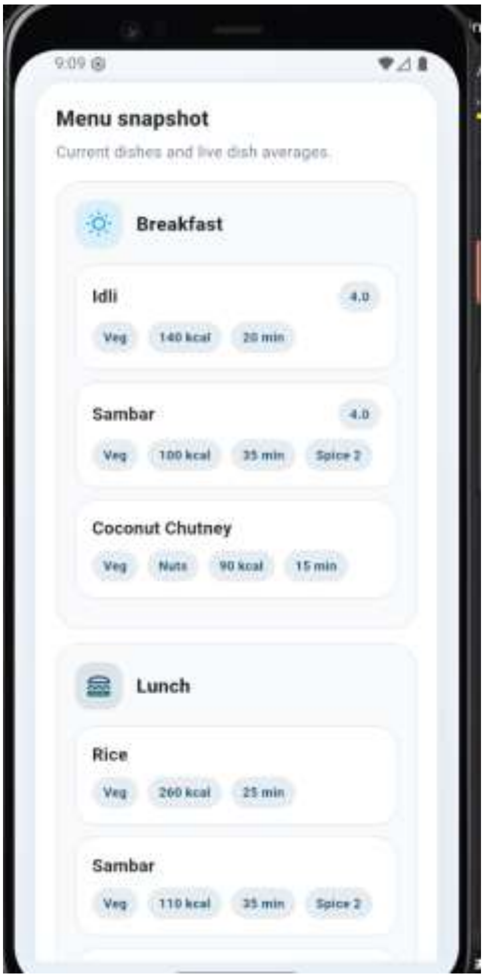

Fig. 3.2.2 — Menu Management Page

Figure 3.2.2 illustrates the admin interface for managing the daily menu, including options to add dishes with nutritional details, set veg/non-veg flags, mark holiday menus, and view menu history.

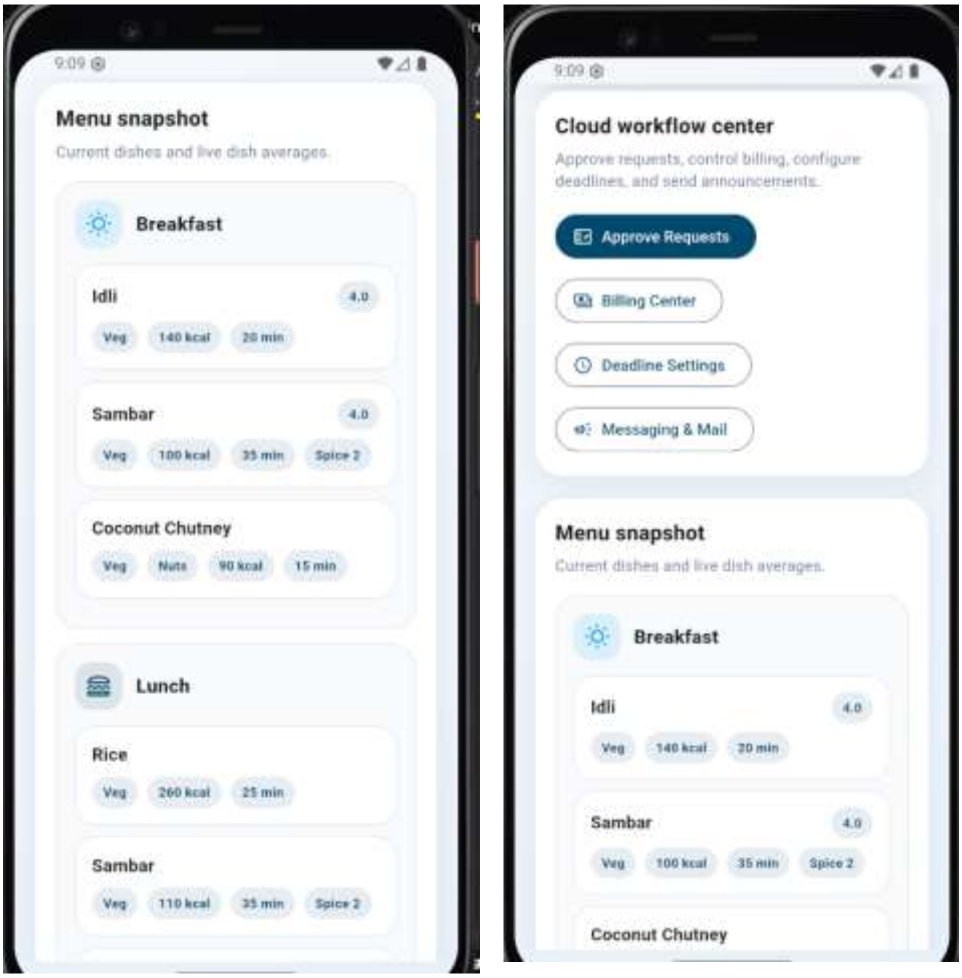

Fig. 3.2.3 — Add Dish Form

Figure 3.2.3 shows the form used by the admin to add new dishes to the daily menu with complete nutritional information including calories, protein content, spice level, prep time, and allergen flags.

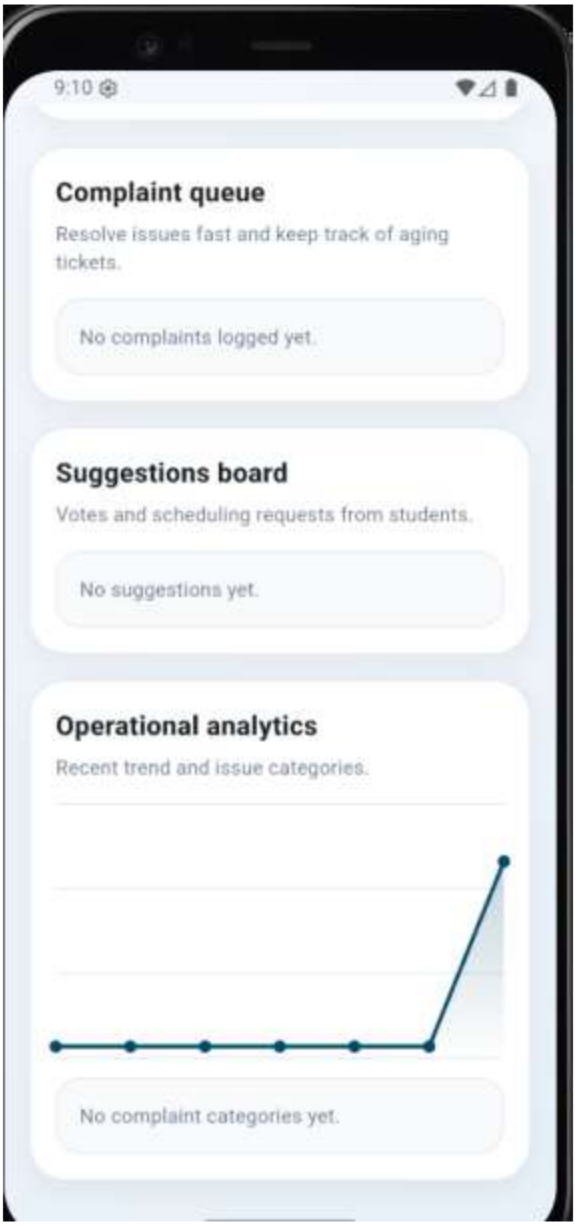

Fig. 3.2.4 — Complaint Management Page

Figure 3.2.4 illustrates the admin complaint management interface for viewing, filtering, assigning, and resolving student complaints. Each complaint displays its severity level, category, and current status for efficient prioritization.

3.3 Student Module

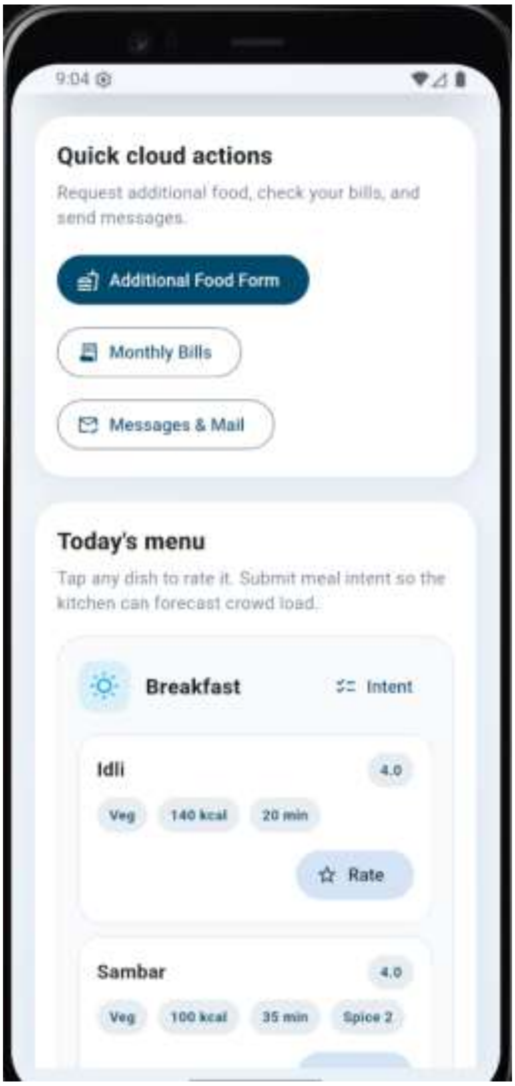

Fig. 3.3.1 — Daily Menu Page

Figure 3.3.1 shows the student interface displaying the daily mess menu organized by meal type. Each dish card shows the name, veg/non-veg indicator, calories, spice level, and preparation time.

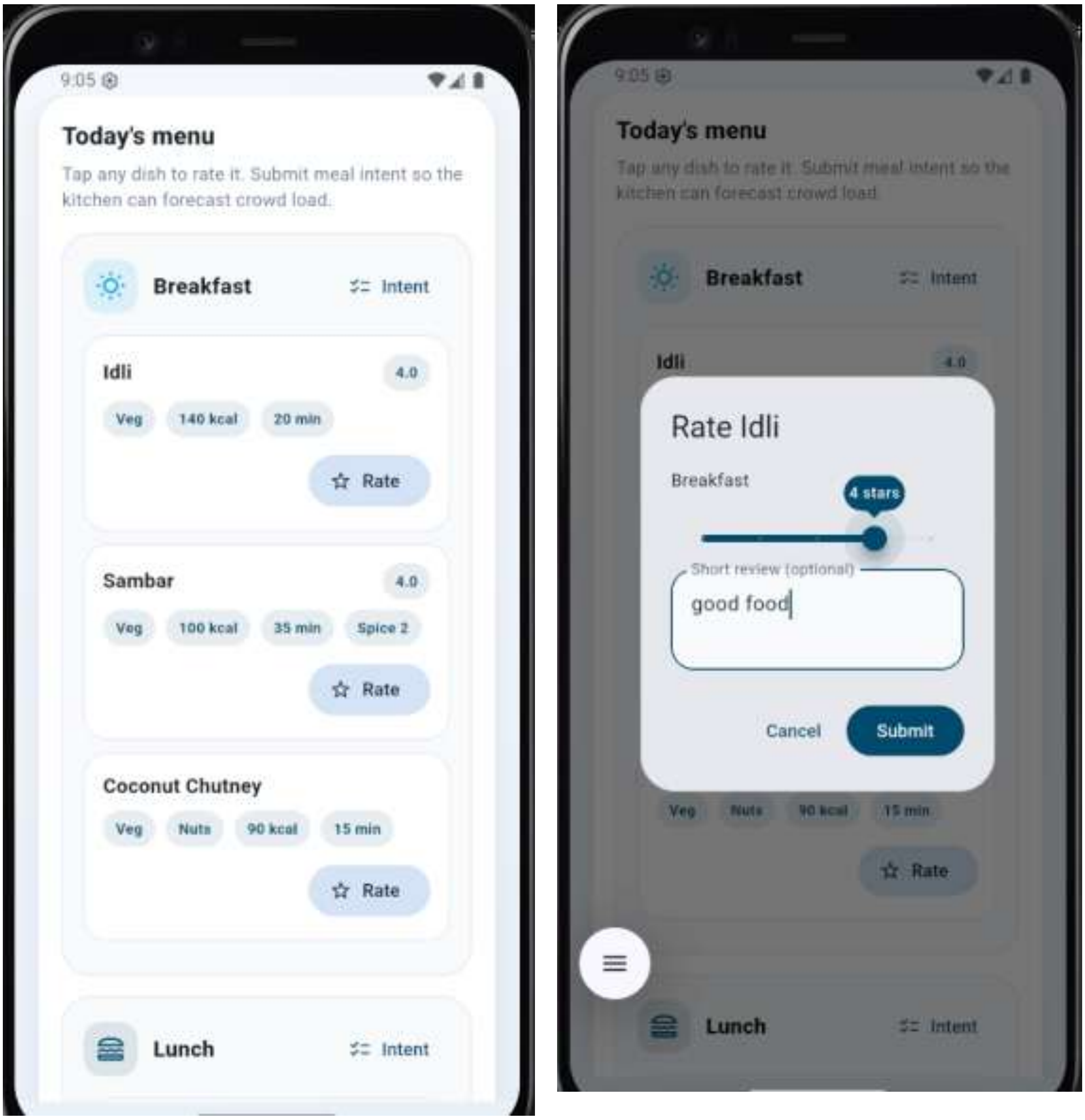

Fig. 3.3.2 — Dish Detail & Rating Page

Figure 3.3.2 illustrates the dish detail and rating screen where students can view complete nutritional information, allergen warnings, and submit star ratings with optional review comments and quality tags such as taste, temperature, and freshness.

Fig. 3.3.3 — Complaint Filing Page

Figure 3.3.3 shows the complaint filing interface where students can submit food quality complaints by selecting a category, describing the issue, setting a severity level, and optionally enabling anonymous submission.

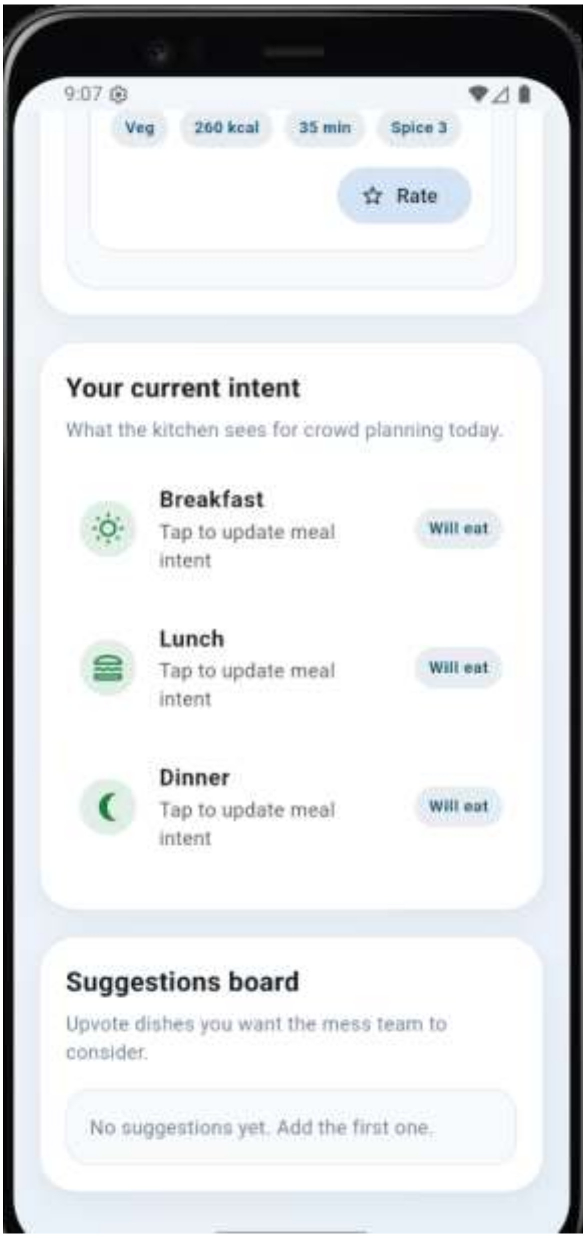

Fig. 3.3.4 — Suggestion Page

Figure 3.3.4 shows the student suggestion interface, allowing students to propose new dishes with preferred meal type, view community-voted suggestions, and upvote existing proposals to support their inclusion in the menu.

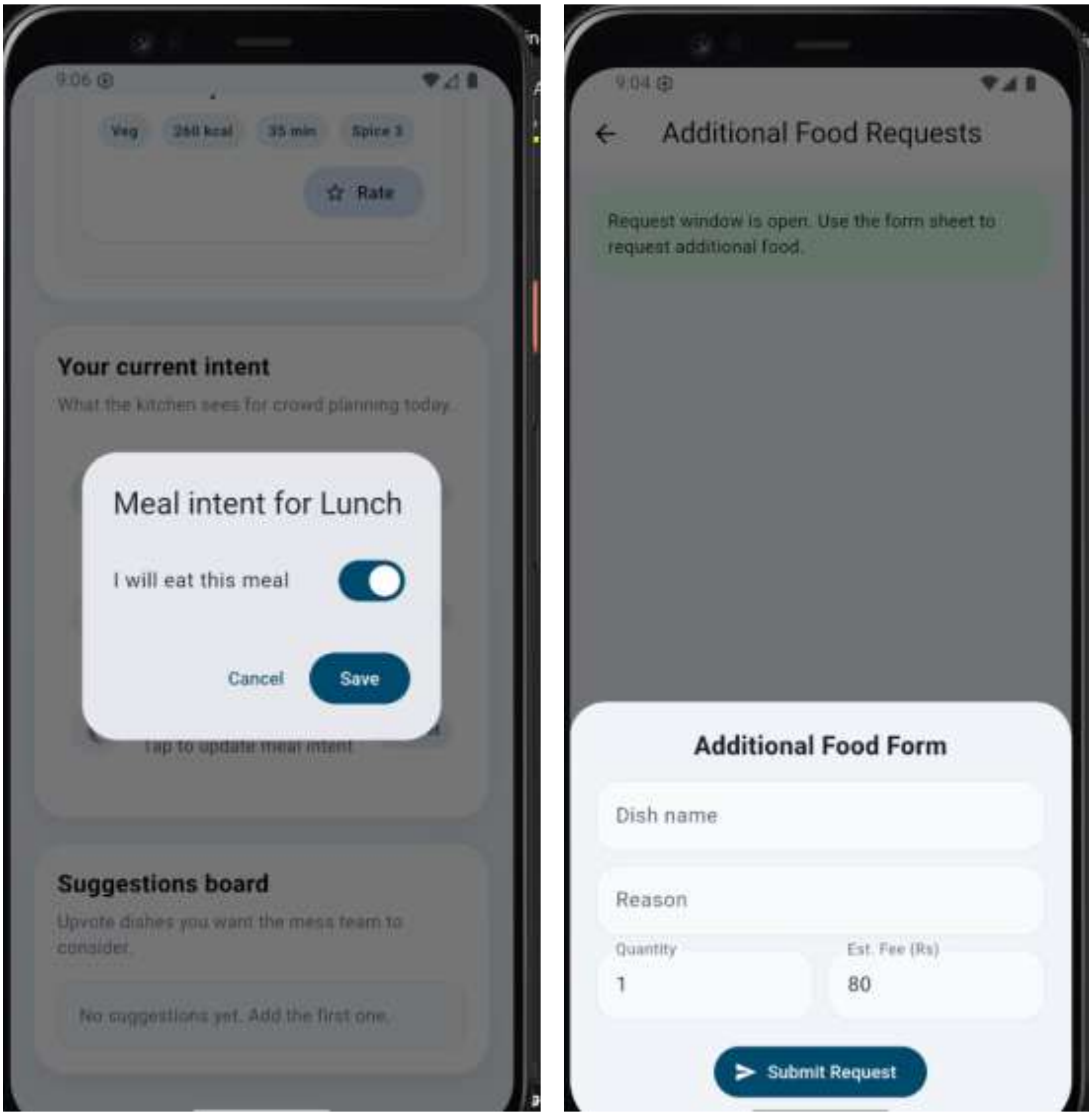

Fig. 3.3.6 — Meal Intent & Additional Request Page

Figure 3.3.6 illustrates the meal intent declaration screen where students indicate their participation for each meal of the day to support accurate food planning, and the additional food request form for ordering extra meals outside the regular schedule.

3.4 Warden Module

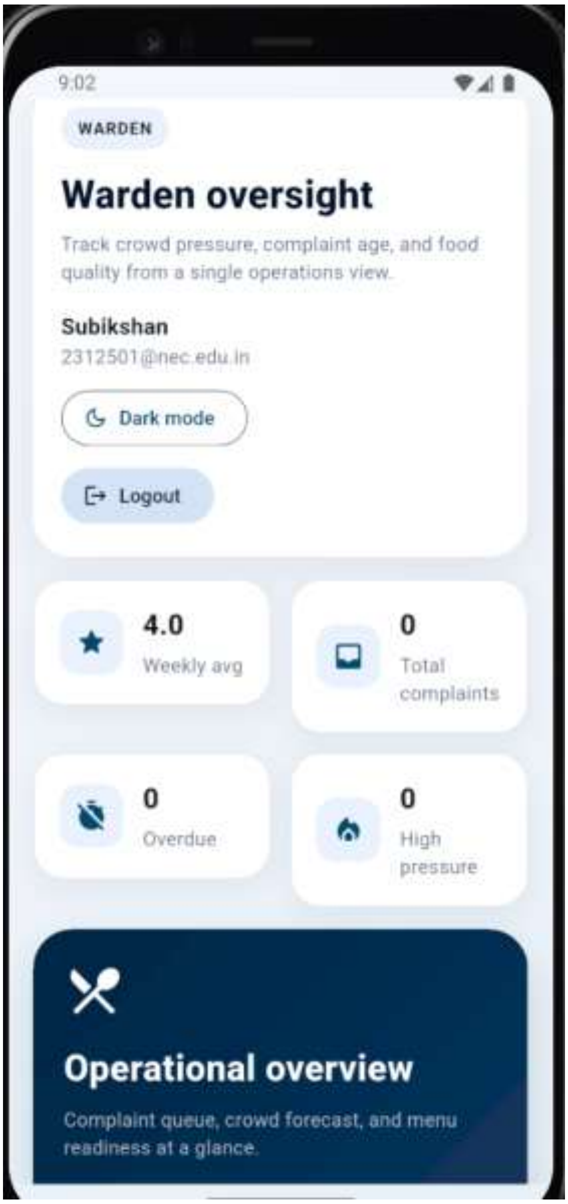

Fig. 3.4.1 — Warden Dashboard

Figure 3.4.1 shows the warden dashboard providing hostel-specific oversight, displaying student activity summaries, daily meal intent statistics, pending complaints from the warden's hostel, and announcement management controls.

CHAPTER 4

CONCLUSION AND FUTURE SCOPE

4.1 CONCLUSION

The Mess Management System successfully demonstrates how traditional college hostel food operations can be transformed into an efficient digital platform using modern technologies. By replacing manual complaint books, verbal feedback, and paper-based billing with an automated mobile application, the system enhances operational efficiency, minimizes communication gaps, and ensures better coordination among students, wardens, and administrators. The integration of Flutter enables a seamless cross-platform experience across Android, iOS, Web, and Desktop from a single codebase. Firebase provides secure authentication, real-time database synchronization, and scalable backend support. Offline functionality through SharedPreferences and SQLite ensures the application remains usable even in low-network hostel environments, which is a critical practical requirement.

The system gives students a structured channel to rate meals, file complaints with tracked resolution, vote on suggestions, and monitor their billing — all in one place. Administrators gain real-time control over menus, complaints, and billing without depending on manual coordination. The role-based architecture ensures that each stakeholder — student, warden, and admin — has a tailored, focused experience. Overall, the Mess Management System increases transparency, boosts service quality accountability, and significantly enhances student satisfaction, making it a reliable and scalable solution for modern college hostel management.

4.2 FUTURE SCOPE

Although the Mess Management System provides a comprehensive digital solution for hostel mess management, several enhancements can be implemented to further improve its functionality and user experience.

Integration of online payment systems will allow students to pay their monthly mess bills securely and conveniently directly through the application, eliminating the need for physical cash transactions. Advanced Firebase Cloud Messaging integration can deliver personalized reminders for meal timings, bill due dates, and complaint status updates in real time.

A dietary preference and health profile feature can allow students to declare allergies and dietary restrictions, enabling the system to flag incompatible dishes and suggest healthier alternatives. Incorporating an AI-based recommendation system can further personalize the menu viewing experience based on individual ratings history and nutritional goals.

Mess attendance analytics and wastage tracking can provide deeper insights into meal participation trends, helping administrators plan food quantities more accurately and reduce waste. Integration with the college's ERP or student information system can automate student registration and billing without requiring manual data entry.

Multi-mess support for campuses with multiple dining facilities can be introduced to manage separate menus, staff, and billing under a single unified platform. Dedicated parent access with periodic meal quality reports and billing notifications can improve transparency with guardians. These future improvements will make the Mess Management System more intelligent, inclusive, and adaptable to the evolving needs of college hostel administration.

LINK: https://github.com/subashmuthub/Campus-Navigation-System.git